



BrainBook

A Stimulating, Personalized Guide for
the Modern Age

Settings

Preferred Learning Style:

☐ Text ☒ Diagrams

Favorite Learning Tools:

- ☐ Games
- ☒ Responses
- ☐ Quizzes
- ☐ Flashcards

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Keywords: Write here... Write here...

Question: Write a new Question here...

Prior Knowledge: Describe what you already know about the topic here...

Open

Existing

AI Feedback

- Integration is the inverse operation of differentiation. It reconstructs a function from its derivative.
 - Let's start with the power rule: What's the power rule for derivatives? (Hint: Start Right to find your response space. Write there?)
- Right! So what if we wanted to go the other way? For example, let's take $y = x^2$. What's the derivative of that using the formula?
- Now, let's go from $2x$ to x^2 . What do you have to do to $2x$ to get to x^2 ?
- Correct. Let's formulate that. We say that $(2x) \cdot x/2$ gets us to x^2 . Now, what if we used n instead of 2 ?

Well, in that case, the formula is the exact same! If we have nx , we divide by n and multiply x : $(nx) \cdot x/n$. Algebraically, that is just equal to x^2 . But what if we just use x ? Can you simplify?

5. Right! But note that x is equal to x^1 , which allow us to generalize for any x^n . If we divide by n and multiply by x :

- $x^n \cdot x/n = x^n(n+1)/n$

This is almost the product rule; however, because we start with some n , we actually use the denominator to $n+1$.

- Product Rule: Integral of $x^n = x^{n+1}/(n+1)$